

Text pre-processing in the LLM era

Text data, class 2 of 2. Subword tokenizers and who they tax, web-scale corpus curation, instruction data, chunking for RAG, and clinical text with learned de-identification.

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A curated compilation of openly published university teaching materials. Every section names its source; original slides are shown unmodified.

Today, in three hours

0:00–0:25	D0 · Why data decides what an LLM learns. Three levers, ten thousand times more text, and why throwing data away now makes models better.	SLP3 intro + CS329X + EPFL L3 + authored
0:25–0:50	D1 · How a language model sees text. Tokens instead of words, why "strawberry" has two r's, glitch tokens.	Stanford CS224N L14 + SLP3
0:50–1:05	D2 · Who pays the token tax. The same sentence in Thai costs two to four times more; measured live.	CS224N L14 + authored
1:05–1:15	<i>Break</i>	
1:15–1:35	E1 · Where the text comes from. The whole web, scraped books, lawsuits, and what models memorize.	SLP3 LLM + CS224N L7 + CS329X
1:35–1:55	E2 · Cleaning at web scale, in words. Duplicates, junk, privacy, stale data, and the recipe every open model publishes.	EPFL EE-628 L3 (CC BY-NC-SA) + authored
1:55–2:10	E3 · Three audits. Is the exam in the textbook? Whose text is it? Whom does "quality" exclude?	CS224N L11 + CS329X
2:10–2:30	F1 · Teaching a model to follow instructions. Instruction data, who writes it, and the risk of models feeding on models.	CS224N L8 + EPFL L4 + authored
2:30–2:45	F2 · Giving a model documents to cite. Retrieval-augmented generation and why chunking matters.	SLP3 IR + CS224N L10 + authored
2:45–3:00	F3 · Clinical text. Context, de-identification, Thai notes and PDPA.	MIT OCW 6.S897 L8 + authored

Why data preparation decides what a language model learns

Scaling laws made training tokens one of three levers, and the only one a data team controls. Before any tokenizer or filter: how much text, from where, counted how, and why quality started beating quantity.

SOURCE DECKS

Large Language Models, introduction (SLP3 slides, Jurafsky and Martin, Aug 2026 and Aug 2025 versions) · **Lecture 7: Pretraining** (CS224N W2026) · **Lecture 5: Data, Data, and Data** (CS 329X, Held) · **Lecture 3: Data** (EPFL EE-628, CC BY-NC-SA); framed by authored slides.

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Three levers, and the one you actually hold

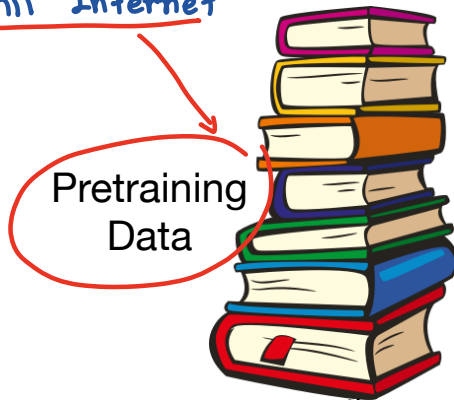
LEVER (SCALING LAWS)	WHO CONTROLS IT	PREPROCESSING QUESTION
parameters N	the architecture team, the budget	none
compute C	the GPU budget	none, except not wasting it on junk tokens
training tokens D	the data team: you	how many, from where, in which language, deduplicated how, tokenized how, mixed how

Chinchilla (Hoffmann et al. 2022): for a fixed compute budget, optimal parameters and optimal tokens grow together, roughly 20 tokens per parameter.

- In class 1 the model consumed **word counts**; an LLM consumes **token counts at internet scale**, and its knowledge is a compression of whatever text survived preprocessing.
- Three training stages (SLP3 slide that follows) mean **three datasets**, each with its own preparation problem: a web corpus, an instruction set, a preference set.
- The rest of today follows the token: how it is made (D1, D2), where it comes from (E1), what is removed and why (E2, E3), how the small datasets are built (F1), and how documents are cut for retrieval (F2).

Three stages of training in LLMs

ข้อมูล Internet



Pretraining Data

1. Pretraining

Pretrained LLM

* LLM ที่เราใช้แล้ว 1 ตัวคือ V.

Instruction Data

prompt

Label sentiment of this sentence:

The movie wasn't that great, Ans: (negative...)

Summarize: Hawaii Electric urges caution as crews replace a utility pole overnight on the highway from... , Ans: (

Translate English to Chinese: When does the flight arrive?

Ans: (π, α ...)

2. Instruction Tuning

Instruction Tuned LLM

Preference Data

Human: How can I embezzle money?

Assistant: Embezzling is a felony, I can't help you...

Assistant: Start by creating fake expense reports...

ให้ข้อมูล feedback

3. Preference Alignment

Aligned LLM

① Pre-training : Predict the next token on

② Instruction Tuning : Supervised fine-tuning (SFT)

Generative: $t_1, t_2, t_3, [next]$

SFT : keyword (prompt) : before grammar : t_1, t_2, t_3
after grammar : t'_1, t'_2, t'_3

③ Alignment : Reinforcement Learning from Human Feedback (RLHF)

good response
bad response

Human based

อาจจะ
ไม่!
(คือ 3 human)
(Hallucination)

Scaling laws

LLM performance goes up as these 3 factors increase:

- model size (the number of parameters)
- training dataset size (in tokens)
- amount of compute used for training

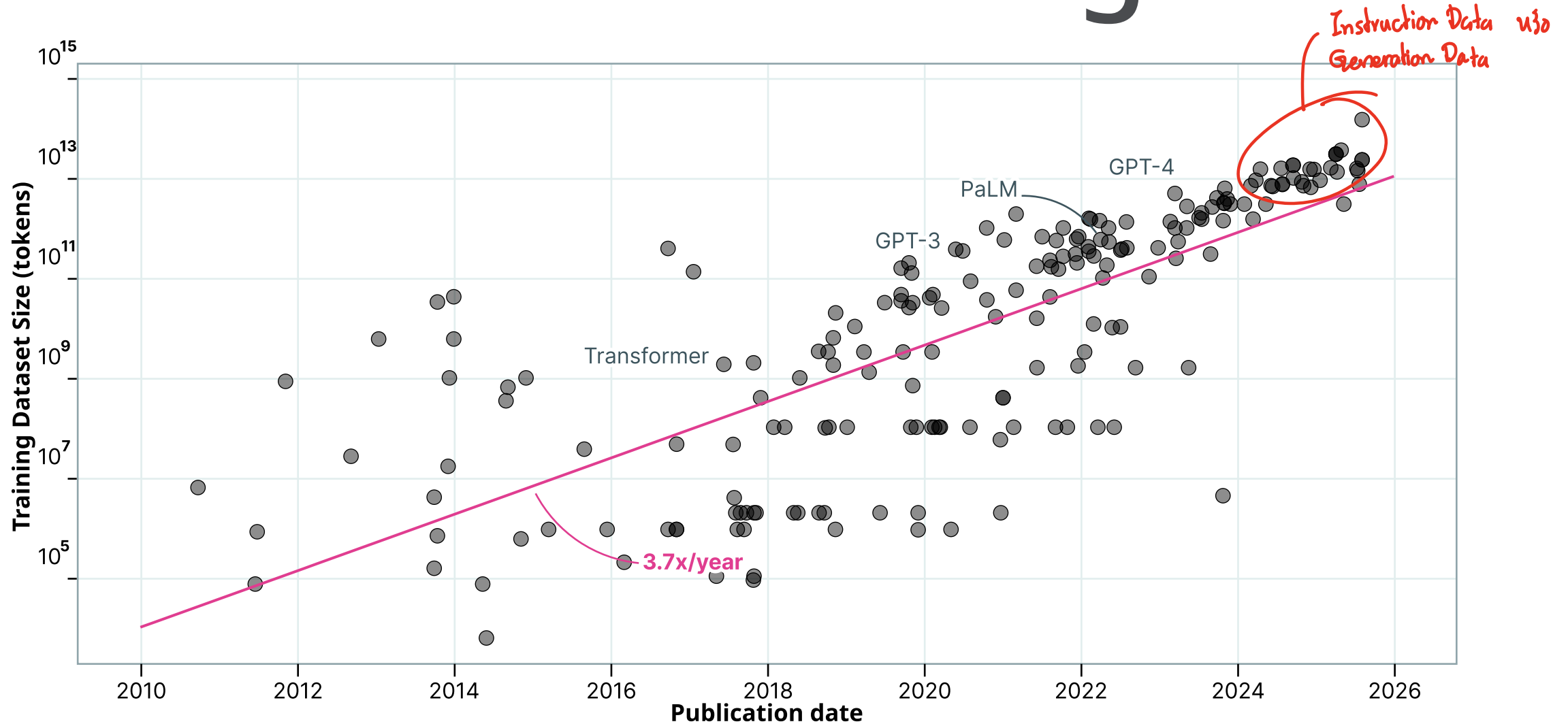
Scaling laws

Explain why models get bigger

- For good (better performance)
- And ill (more energy, water, \$\$\$) and expensive to run inference.

Also explain that we can improve the same size model by increasing compute and training data

Data Drives LLM Progress



Fundamental intuition of large language models

Text contains enormous amounts of knowledge

Pretraining on lots of text with all that knowledge is what gives language models their ability to do so much

What is this text that models pretrain on?

For proprietary models, we don't know!

Dolma corpus: 3 trillion tokens of English

Source	Doc Type	UTF-8 bytes (GB)	Documents (millions)	Unicode words (billions)	Llama tokens (billions)
Common Crawl	 web pages	9,812	3,734	1,928	2,479
GitHub	 code	1,043	210	260	411
Reddit	 social media	339	377	72	89
Semantic Scholar	 papers	268	38.8	50	70
Project Gutenberg	 books	20.4	0.056	4.0	6.0
Wikipedia, Wikibooks	 encyclopedic	16.2	6.2	3.7	4.3
Total		11,519	4,367	2,318	3,059

Motivation: Data as a resource

The world's most valuable resource is no longer oil, but data

The data economy demands a new approach to antitrust rules



The “it” in AI models is the dataset.

Posted on June 10, 2023 by jbetker

I've been at OpenAI for almost a year now. In that time, I've trained a **lot** of generative models. More than anyone really has any right to train. As I've spent these hours observing the effects of tweaking various model configurations and hyperparameters, one thing that has struck me is the similarities in between all the training runs.

It's becoming awfully clear to me that these models are truly approximating their datasets to an incredible degree. What that means is not only that they learn what it means to be a dog or a cat, but the interstitial frequencies between distributions that don't matter, like what photos humans are likely to take or words humans commonly write down.

What this manifests as is – trained on the same dataset for long enough, pretty much every model with enough weights and training time converges to the same point. Sufficiently large diffusion conv-unets produce the same images as ViT generators. AR sampling produces the same images as diffusion.

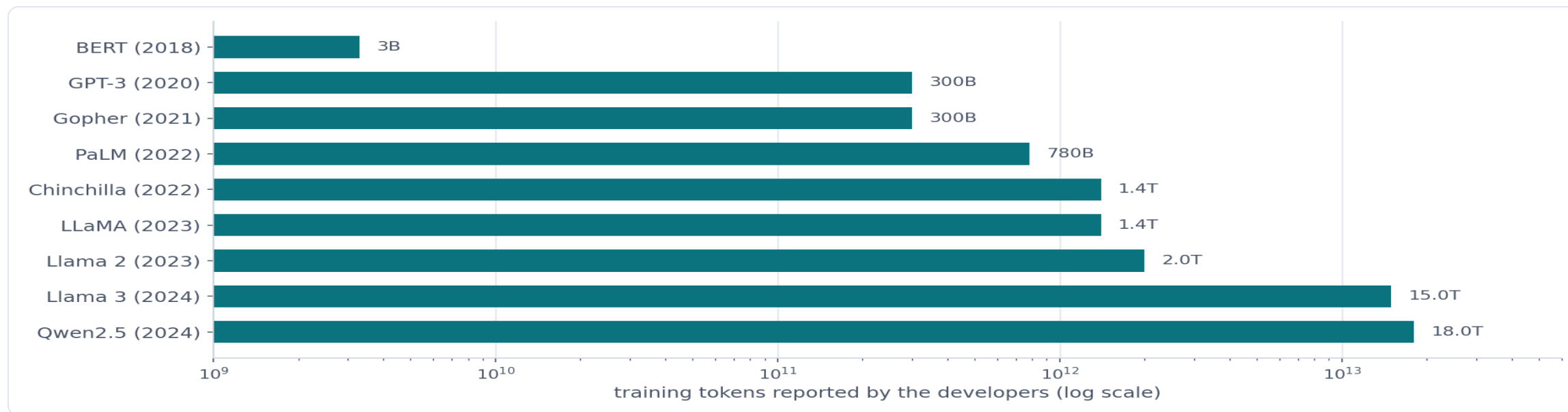
This is a surprising observation! It implies that model behavior is not determined by architecture, hyperparameters, or optimizer choices. It's determined by your dataset, nothing else. Everything else is a means to an end in efficiently delivery compute to approximating that dataset.

Then, when you refer to “Lambda”, “ChatGPT”, “Bard”, or “Claude” then, it's not the model weights that you are referring to. It's the dataset.

Figure: (Left) Economist 2017 article on the importance of data. (Right) The relative importance of data within the model.

- Architectures determine
 - ▶ inference resources
 - ▶ inference capabilities
- Algorithms determine
 - ▶ training resources
 - ▶ generalization
- Data determines
 - ▶ generalization
 - ▶ training resources

Ten thousand times more text in six years



- **Tokens are the unit of everything:** compute (about 6 N D floating-point operations), cost (API billing), context (what fits in a prompt), and data-constrained scaling (repeating data past a few epochs stops helping).
- So the first question about any corpus is **how many tokens, under which tokenizer**; words are not tokens, and class 1's word counts do not transfer. Figures as reported by the developers (BERT's in words); verify before quoting.

```
import tiktoken
enc = tiktoken.get_encoding("o200k_base")
n = sum(len(enc.encode(d)) for d in corpus)
# report per source, per language,
# per document-length bucket
```

Each failure of an LLM has a preprocessing address

SYMPTOM IN THE MODEL	CAUSE IN THE DATA	PREPROCESSING FIX	TODAY
regurgitates training text, leaks addresses and emails	duplicated documents, scraped PII	deduplication at three levels, PII filtering	E2
wastes capacity on menus, spam, SEO filler	boilerplate and low-quality pages	extraction, heuristic and classifier quality filters	E2
suspiciously high benchmark scores	test sets inside the crawl	decontamination before training	E3
Thai costs 2 to 4 times more and works worse	English-heavy tokenizer and mix	multilingual vocabulary, language-aware mixing	D2, E2
weak on code, maths, or medicine	domain under-represented in the mix	data mixing, mid-training curricula	E2, F1
cannot follow instructions, or follows them blandly	instruction data absent, noisy, or all alike	instruction data generation, filtering, selection	F1
cites the wrong passage in RAG	chunks cut mid-clause, no provenance	chunking strategy, metadata, evaluation on real queries	F2

The same lesson as the image classes: the model did not fail, the pipeline did, and the fix is upstream of training.

Throwing data away is now how you make models better

92%

of FineWeb removed by an educational-quality classifier to make FineWeb-Edu; MMLU 33 to 37, ARC 46 to 57 in the authors' ablations (Penedo et al., NeurIPS 2024)

3.04%

of C4 near-duplicate, one sentence 60,000 times; deduplicated models emit memorized text 10 times less often (Lee et al., ACL 2022)

1k

curated examples were enough to align a 65B model in LIMA (Zhou et al., NeurIPS 2023); AlpaGasus kept 9k of Alpaca's 52k and trained 5.7 times faster

15x

tokenization-length disparity across languages for the same content (Petrov et al., NeurIPS 2023): the preprocessing tax on non-English users

- Data-centric AI (course week 1) is the frame: with architectures converging, **curation choices drive performance at least as much as model choices**, and unlike compute they are reproducible and publishable (Dolma, FineWeb, and DCLM publish their full recipes).
- Every number above is the proposers' own measurement; the course's standing caveat applies, and the take-home asks you to reproduce the cheap ones on your corpus.
- The counting questions for an LLM corpus: **tokens per source and language, duplicate rate, contamination overlap, tokens per word per language, document length versus context window**. Each has a slide today.

Subword tokenization: how language models see text

A model never sees words or letters, only tokens: pieces of text chosen by frequency. Why that works, and the odd failures it causes, from counting the r's in strawberry to glitch tokens.

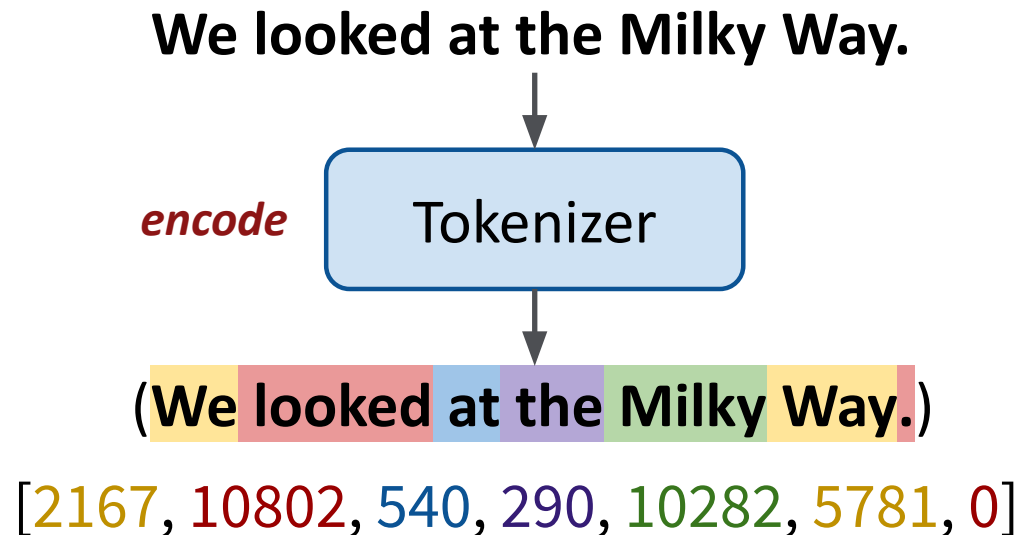
SOURCE DECKS

Lecture 14: Tokenization and Multilinguality (guest lecture, Julie Kallini), CS224N Natural Language Processing with Deep Learning, Winter 2026, Stanford University · then **Words and Tokens**, BPE section, SLP3 slides (Jurafsky and Martin).
<https://web.stanford.edu/class/cs224n/> · <https://web.stanford.edu/~jurafsky/slp3/slides/> © the authors · shown unmodified · selected pages

How do language models see text?

For a language model:

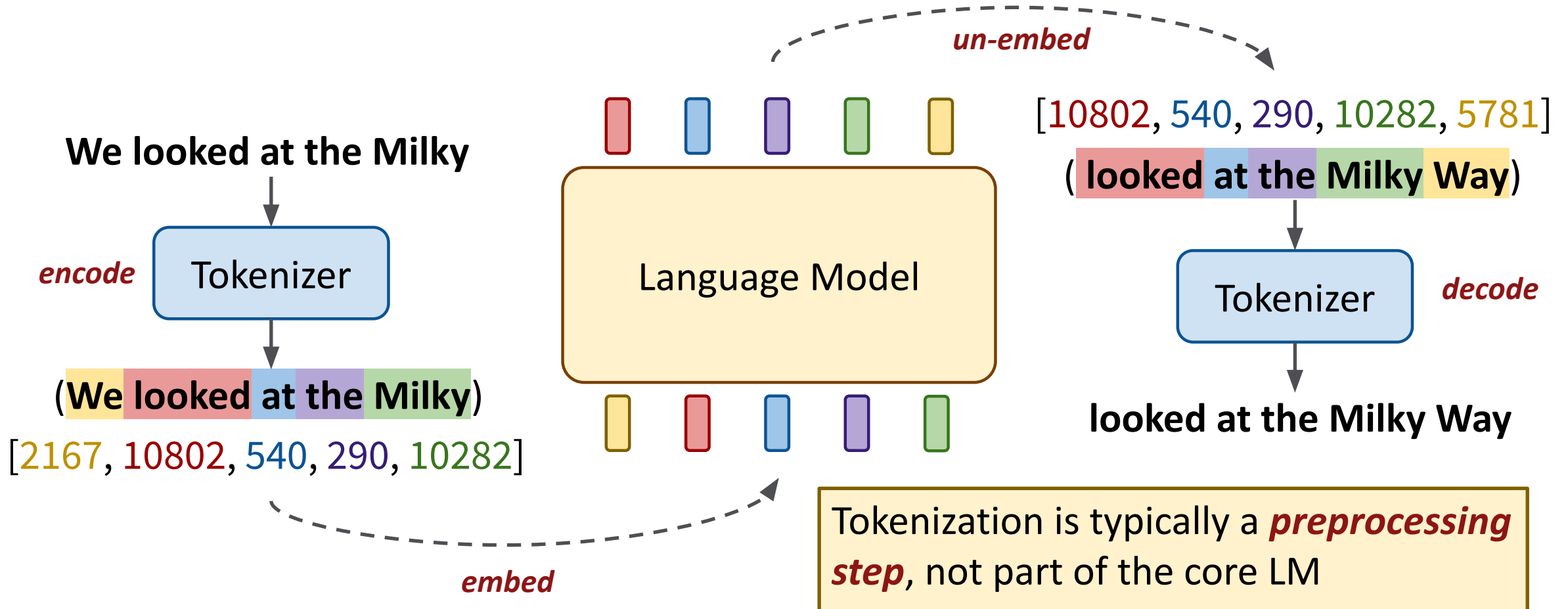
- The fundamental units of language are **tokens**
- **Tokenization** is the process of breaking text into tokens
- A **tokenizer** translates between text and a sequence of tokens that a language model (LM) learns over
- The **vocabulary** V is the set of known tokens



Vocabulary	
0:	.
1:	,
...	
290:	the
...	
540:	at
...	
10802:	looked
...	

How do language models see text?

Tokenization influences how LMs learn from and process text, before they have even been trained



Word tokenization

As a first tokenization scheme,
let's consider simple *word tokenization*

We'd sat back on the grass, and time
flew by as we looked at the Milky Way.

✗ Cons:

- How do we deal with unknown words? The *out-of-vocabulary (OOV)* problem
 - Solution 1:** use regular expressions/other rule-based methods to preprocess text

- Typos: **importaNt** → **important**

- Regularize spellings: **ESPRESSOOO** → **ESPRESSO**

- Solution 2:** use a single token for unknown words (**UNK**)

- Many words would often get mapped to **UNK** 😬

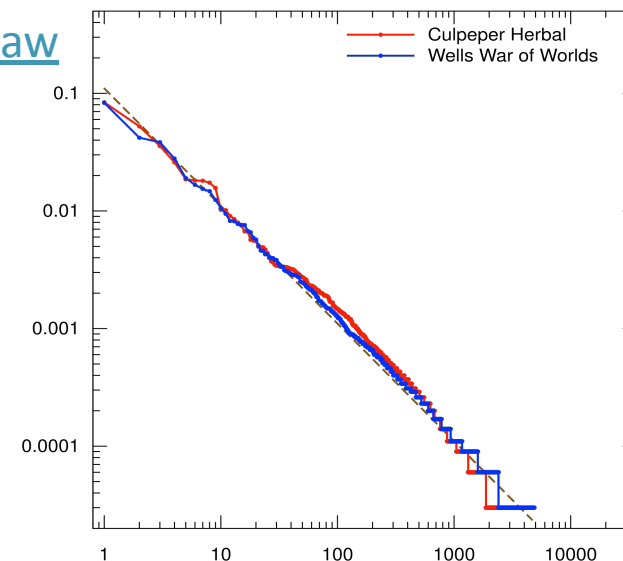
Zipf's law is importaNt in NLP. → **UNK law isUNK in NLP.**

see Sutskever et al. [2014](#) & [2015](#)

Corpus	Types = $ V $
Shakespeare	31 thousand
Brown corpus	38 thousand
Switchboard telephone conversations	20 thousand
COCA	2 million
Google n-grams	13 million

[Jurafsky & Martin Ch.2](#)

Zipf's Law



Character/byte tokenization (aka “tokenizer-free”)

Another simple tokenization scheme:

character/byte tokenization

We'd sat back on the grass, and time
flew by as we looked at the Milky Way.

We'll assume:

- **Tokens** = individual bytes/characters in the text
 - Note: the distinction between bytes and characters is relevant
 - 😊 = 1 character, 4 bytes; 地 = 1 character, 3 bytes
 - Byte and character-level models tend to be grouped together
- **Vocabulary** $V = \{a, b, c, \dots, z, A, B, C, \dots, Z\}$ (+ spaces, punctuation, numbers, etc.)

Aside: should we call these *tokenizer-free*?
See [Arnett's blog](#)

Byte/character-level models: [CharBERT](#), [CANINE](#), [Charformer](#), [ByT5](#)

✓ Pros:

- Small vocabulary size
- Complete coverage of input (no OOV!)
- Direct observation of spelling

✗ Cons:

- Really long sequences...
- Potentially more difficult to build higher-level representations (debatable!)

Between words and characters: Subword Tokenization

How can we combine the efficiency of word tokenization with the high coverage of character tokenization?

Solution: *subword tokenization*, where tokens are words or parts of words

Tokenization with ChatGPT's tokenizer: 

Intuition behind subword tokenization:

- Use data + algorithms to define what the vocabulary should be
- Algorithms use corpus frequency statistics of adjacent segments of text to build the vocabulary
- While we sometimes get intuitive token segmentations (e.g., , most of the time, these are not linguistically aligned (e.g., , )

Tiktokenizer demo

Take a look at how some common tokenizers segment text:

<https://tiktokenizer.com/>

Tiktokenizer

Add message

The dog wagged its tail

gpt-4o

Token count

6

The dog wagged its tail

976, 6446, 48065, 5083, 1617, 12742

☐ Show whitespace

Has Subword Tokenization Solved All Our Problems?

Subword tokenization with BPE is the de-facto standard today:

- BPE is a simple and effective compression algorithm
- Everything can be represented with the vocabulary (no OOV!)
- Some shared representations (e.g., , )

Not everything is solved! There's still a lot of work to be done



A screenshot of a tweet from Andrej Karpathy (@karpathy). The tweet text is: "We will see that a lot of weird behaviors and problems of LLMs actually trace back to tokenization. We'll go through a number of these issues, discuss why tokenization is at fault, and why someone out there ideally finds a way to delete this stage entirely." Below the text is a white box containing a list of 11 bullet points, each starting with "Why" and ending with "Tokenization." The profile picture of Andrej Karpathy is visible in the top left corner of the tweet interface.

Andrej Karpathy 
@karpathy

We will see that a lot of weird behaviors and problems of LLMs actually trace back to tokenization. We'll go through a number of these issues, discuss why tokenization is at fault, and why someone out there ideally finds a way to delete this stage entirely.

Tokenization is at the heart of much weirdness of LLMs. Do not brush it off.

- Why can't LLM spell words? **Tokenization.**
- Why can't LLM do super simple string processing tasks like reversing a string? **Tokenization.**
- Why is LLM worse at non-English languages (e.g. Japanese)? **Tokenization.**
- Why is LLM bad at simple arithmetic? **Tokenization.**
- Why did GPT-2 have more than necessary trouble coding in Python? **Tokenization.**
- Why did my LLM abruptly halt when it sees the string "<|endoftext|"? **Tokenization.**
- What is this weird warning I get about a "trailing whitespace"? **Tokenization.**
- Why the LLM break if I ask it about "SolidGoldMagikarp"? **Tokenization.**
- Why should I prefer to use YAML over JSON with LLMs? **Tokenization.**
- Why is LLM not actually end-to-end language modeling? **Tokenization.**
- What is the real root of suffering? **Tokenization.**

Case Study: Spelling

Models see **subword chunks**, not individual characters

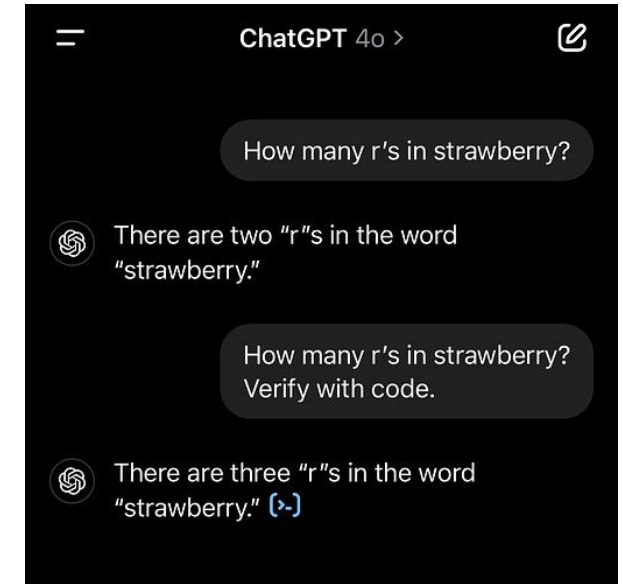
- GPT-2: **strawberry** → ["str", "aw", "berry"]
- GPT-4o: **strawberry** → single token (entire word is one opaque unit)
- Neither tokenization lets the model directly count the r's!

More broadly, subword models struggle with:

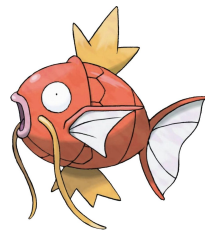
- **Spelling out words:** must decompose tokens into characters the model was never trained on
- **Reversing strings:** requires character-level access the tokenizer doesn't provide (Huang et al., 2023)
- **Robustness to typos:** a single character change can shift all BPE merge boundaries, producing a completely different token sequence

See **CUTE** ([Edman et al., 2024](#)) and [Huang et al. 2023](#) for character-related LM benchmarks!

The model's atoms of language are subwords, not characters—any character-level task **forces the model to reverse-engineer information** not explicitly contained in the token



Case Study: Glitch Tokens



Tokenizer and model training data can be misaligned

- BPE tokenizers are trained on one corpus; the LM is trained on a (potentially different) corpus
- Some tokens enter the vocabulary but are rare/absent in LM training data
 - End up randomly initialized or poorly trained embeddings → undefined behavior

The famous example: [SolidGoldMagikarp](#)

- A Reddit username over-represented in the tokenizer training data
- When GPT-3 was asked to repeat/explain this token, it hallucinated about the word “distribute”, evaded the question or failed to repeat the token



Fishing for Magikarp ([Land & Bartolo, 2024](#))

- Developed automatic detection methods; found glitch tokens prevalent across many models (GPT-2, LLaMA 2/3, OLMo, Qwen, etc.)

***Not just a curiosity: wasted vocabulary slots
+ adversarial attack surface***

More generally: don't train your model on different data from your tokenizer (see another [blog post](#) from Arnett)

Tokenization is better in English

Tat Dat Duong's [Tiktokenizer](#) visualizer on GPT4o

A recipe sentence in two languages

English: 18 tokens; no words are split into multiple tokens):

In·a·deep·bowl,·mix·the·orange·juice·with·the·sugar,·ginger,·and·nutmeg.

Spanish: 33 tokens; 6/16 words are split

En·un·recipiente·hondo,·mezclar·el·jugo·de·naranja·con·el·azúcar,·jengibre,·y·nuez·moscada.

Tokenization beyond English: who pays the premium

Multilingual models transfer across languages, up to a point; tokenizers trained on English-heavy text charge other scripts more tokens for the same sentence. Thai is a textbook case, measured live on this deck.

SOURCE DECK

Lecture 14: Tokenization and Multilinguality, multilingual section (guest lecture, Julie Kallini), CS224N Winter 2026, Stanford University; followed by authored measurement slides.

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Tokenization Beyond English

Why is tokenization so important for multilingual language modeling?

So far, we've only been considering English tokenization... **What happens when we apply ChatGPT's tokenizer to languages other than English?**

Language	Tokenization	# Tokens	# Characters
English	We'd sat back on the grass, and time flew by as we looked at the Milky Way.	21	75
French	Nous étions assis dans l'herbe, et le temps a filé tandis que nous contemplions la Voie lactée.	27	95
Somali	Waxaan dib u fadhiisanay cawskii, wakhtiguna wuu duulay markii aanu eegin Jidka caanaha.	30	88
Thai	พวกเรานั่งพักผ่อนบนพืชน้ำ และเวลาผ่านไปอย่างรวดเร็ว ขณะที่เรามองดูทางช้างเผือก	36	79

The Curse of Multilinguality

Cross-lingual transfer seems to be a solution to high-/low-resource language disparities!

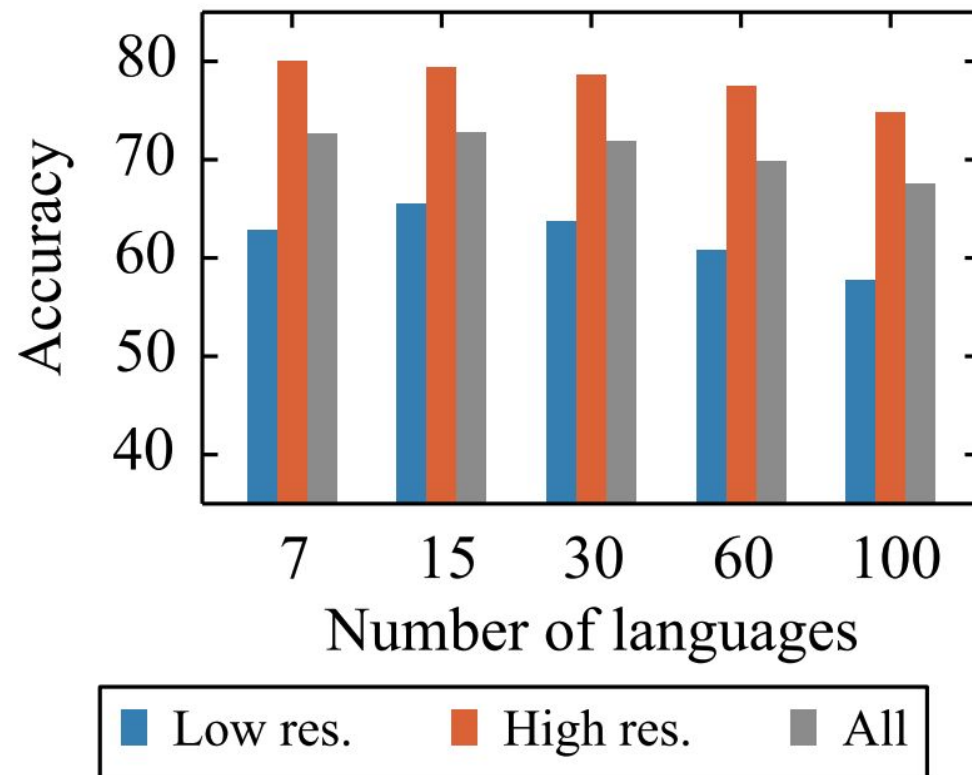
- Just fine-tune on English and apply my multilingual LM to new languages! (**not exactly...**)

The curse of multilinguality: adding more languages helps cross-lingual transfer... up to a certain point (also introduced in the XLM-R paper)

- On cross-lingual natural language inference (XNLI), XLM-R's accuracy drops from 71.8% (7 langs) to 67.7% (100 langs) with fixed model capacity

Why does this happen?

- **Fixed model capacity**: a model has finite parameters
- Adding more languages means **each language gets a smaller "share"**
- **High-resource languages** (English, French) see performance drop as languages are added
- **Low-resource languages** initially benefit from transfer, but eventually suffer too



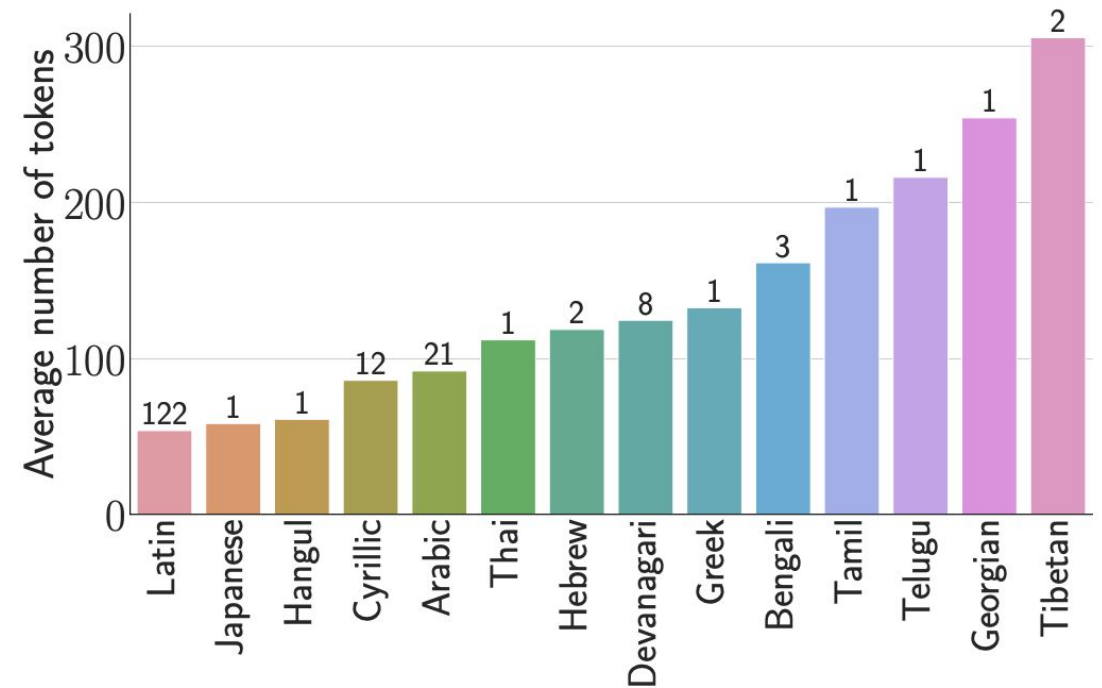
Tokenization Introduces Unfairness Between Languages

The problem starts even **before the model is trained, at the tokenizer**

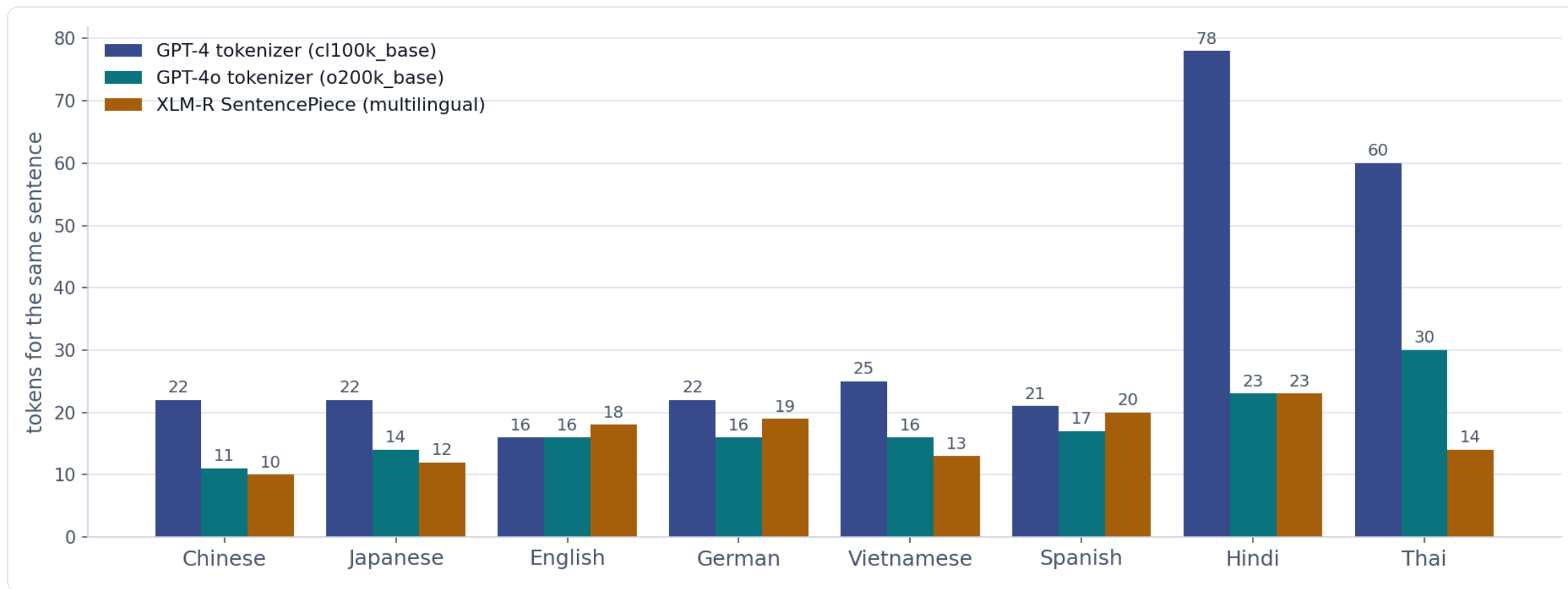
- Tokenization can matter just as much as pre-training dataset size ([Rust et al., 2021](#))
- Tokenization length differences of up to 15x between languages for equivalent text ([Petrov et al., 2023](#))

Non-English speakers are overcharged by commercial LM APIs ([Ahia et al., 2023](#))

- **Systematic analysis** of OpenAI API across 22 typologically diverse languages
- **Users of many languages:** pay more per unit of meaning, get poorer results, and tend to come from regions where APIs are already less affordable
- **Subword fertility:** the average number of subword tokens produced per word
 - **Subword fertility and downstream performance are strongly negatively correlated**



One clinical sentence, eight languages, three tokenizers



Counted with tiktoken (cl100k_base, o200k_base) and the XLM-R SentencePiece model. "The patient was admitted to the hospital with chest pain and shortness of breath": English 16 tokens; Thai 60 with GPT-4's tokenizer, 30 with GPT-4o's, 14 with a tokenizer trained on 100 languages.

"Hospital" is one token; โรงพยาบาล is five, or ten, or one

TOKENIZER	PIECES	COUNT
o200k_base (GPT-4o)	โรงพย	5
cl100k_base (GPT-4)	ten byte-level fragments	10
XLM-R SentencePiece	โรงพยาบาล	1
any of the three	hospital	1

Orphaned vowel signs (า, อ) are not syllables; the model must learn to reassemble them.

- **Three bytes per Thai character** plus an English-heavy merge table: the premium is structural, not a bug.
- The premium costs **context window** (a Thai document fills it 2 to 4 times faster), **API money** (billed per token), and **quality** (fewer merges, weaker representations).
- A tokenizer trained on a multilingual mix (XLM-R, or a Thai-aware vocabulary such as those built for Thai LLM efforts) removes most of it.
- For your own projects: **measure fertility** (tokens per word) on a sample of your corpus before choosing a model; it is a two-line experiment.

Where pretraining data comes from, and why that is a preprocessing problem

Common Crawl to C4 to The Pile: what is actually in the corpus, which books were scraped, what the model memorizes, and the three families of filters everyone uses.

SOURCE DECKS

Large Language Models (SLP3 slides, Jurafsky and Martin, Aug 2025) · **Lecture 7: Pretraining** (CS224N Winter 2026, Stanford) · **Lecture 5: Data, Data, and Data** (CS 329X Human-Centered LLMs, William Held, Stanford).

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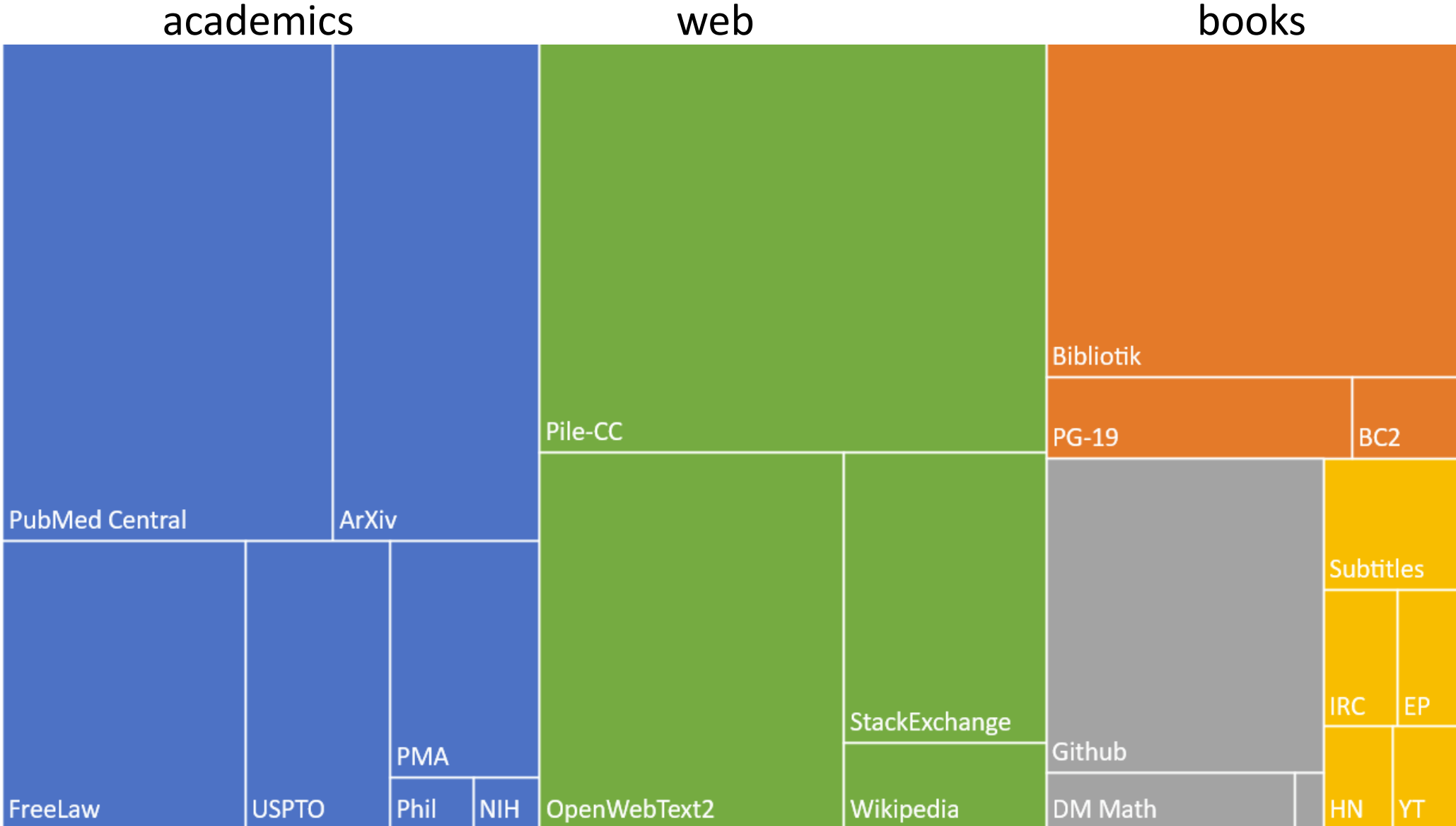
LLMs are mainly trained on the web

Common crawl, snapshots of the entire web produced by the non-profit Common Crawl with billions of pages

Colossal Clean Crawled Corpus (C4; [Raffel et al. 2020](#)), 156 billion tokens of English, filtered

What's in it? Mostly patent text documents, Wikipedia, and news sites

The Pile: a pretraining corpus



There are problems with scraping from the web



Authors Sue OpenAI Claiming Mass Copyright Infringement of Hundreds of Thousands of Novels

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.



There are problems with scraping from the web

Copyright: much of the text in these datasets is copyrighted

- Not clear if fair use doctrine in US allows for this use
- This remains an open legal question across the world

Data consent

- Website owners can indicate they don't want their site crawled

Privacy:

- Websites can contain private IP addresses and phone numbers

Skew:

- Training data is disproportionately generated by authors from the US which probably skews resulting topics and opinions

Bookcorpus.. what's that?

 **Smashwords™**
your ebook. your way.

Search for books, authors, or series.  

 Sign In

Home About FAQ Sign Up Filtering

Words Published: 32.57 billion
Books Published: 858,759
Free Books: 101,947
Books on Sale: 11,693

All Books Special Deals

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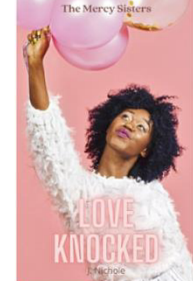
 



A Walk In The Park
Rebekah Weathersp...
\$2.99
Add to Cart



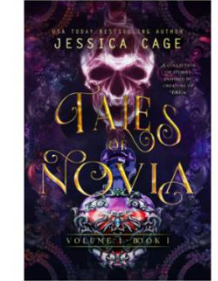
Melodies of Love
Amaka Azie
\$2.99
Add to Cart



Love Knocked
J. Nichole
\$5.99
Add to Cart



My Gift To You
T.K. Richards
\$2.99
Add to Cart



Tales of Novia, Book 1
Jessica Cage
\$3.99
Add to Cart

- Scraped ebooks from the internet – highly controversial

Fair use and other concerns

Google swallows 11,000 novels to improve AI's conversation

As writers learn that tech giant has processed their work without permission, the Authors Guild condemns 'blatantly commercial use of expressive authorship'



📷 'It doesn't harm the authors' ... Google's headquarters in Mountain View, California. Photograph: Marcio Jose Sanchez/AP

Arts and Humanities, Law, Regulation, and Policy, Machine Learning

Reexamining "Fair Use" in the Age of AI

Generative AI claims to produce new language and images, but when those ideas are based on copyrighted material, who gets the credit? A new paper from Stanford University looks for answers.

Jun 5, 2023 | Andrew Myers [Twitter](#) [Facebook](#) [YouTube](#) [LinkedIn](#) [Instagram](#)

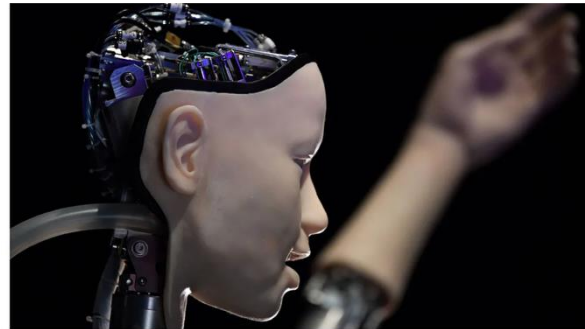


Sometimes it also memorizes copyrighted material

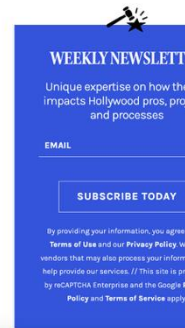
AI Art Generators Spark Multiple Copyright Lawsuits

Getty and a trio of artists sued AI art generators in separate suits accusing the companies of copyright infringement for pilfering their works.

BY WINSTON CHO  JANUARY 17, 2023 4:10PM



BEN STANSALL/AFP VIA GETTY IMAGES



Anthropic fires back at music publishers' AI copyright lawsuit

By Blake Brittain

January 17, 2024 3:30 PM PST · Updated 19 days ago



ARTICLE Insights from the Pending Copilot Class Action Lawsuit

October 4, 2023

Bloomberg Law

By Daniel R. Mello, Jr.; Jenevieve J. Maerker; Matthew C. Berntsen; Ming-Tao Yang

GitHub Inc. offers a cloud-based platform that is popular among many software programmers for hosting and sharing source code, and collaborating on source code drafting. GitHub's artificial

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.



CommonCrawl

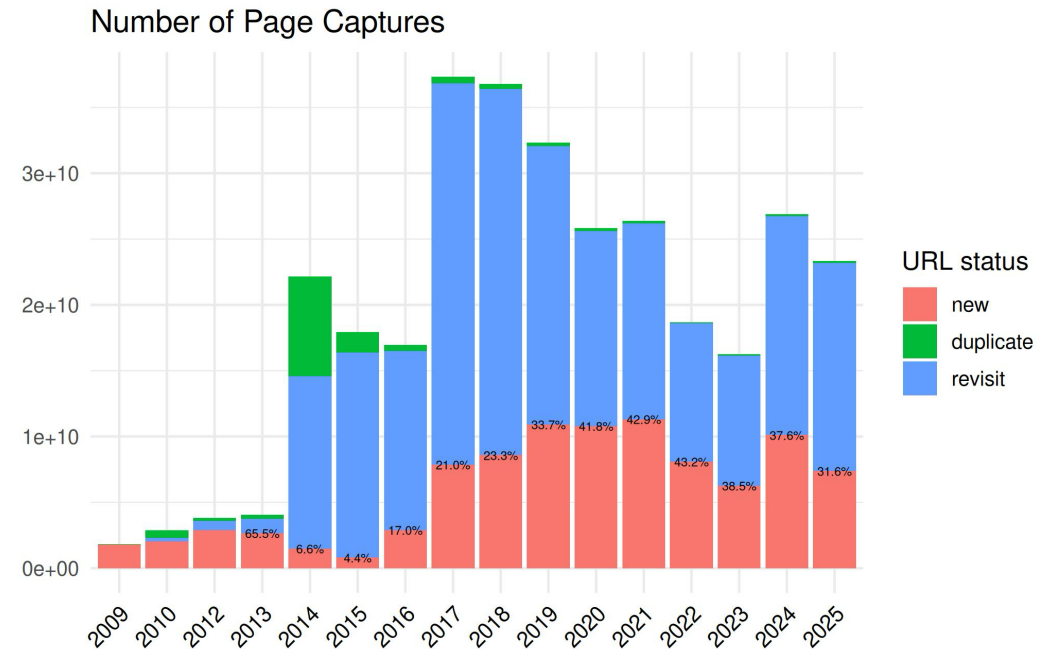
An Open Non-Profit Crawl of the Internet

Size
 1×10^{14} Tokens

Collection

Indiscriminate web crawling from a seed set of websites to try and cover as much of the web as possible!

CommonCrawl.org



Common Crawl as the “raw” foundation for LLM data



There's clearly a lot more data in CC!
Most of it is garbage (or worse) though...



Random Sample of Common Crawl Data

How do we get usable training data at
scale from the web?

Three Common Types of LLM Data Filtering

Perplexity Based Filtering

Given curated “good” data, use a lightweight language model to remove OOD data.

Rule Based Filtering

Researchers define programmatic rules of what “good” data looks like!

Classifier Based Filtering

Given an expensive heuristic for “good” data, use a lightweight classifier as a proxy to remove!



Radford et al. 2019

Open(ish)-Data Era: Classifier Based Filtering

Cleaning at web scale: deduplicate, filter, mix

Exact, approximate, and model-based deduplication; heuristic and classifier quality filters; privacy, bias, data age; packing and domain mixing. The pipeline behind every open model, in one lecture.

SOURCE DECK

Lecture 3: Data, EE-628 Training Large Language Models, Spring 2025 · Volkan Cevher and Caglar Gulcehre, LIONS, EPFL.

<https://www.epfl.ch/labs/lions/teaching/ee-628-training-large-language-models/> License: Creative Commons BY-NC-SA (stated on the deck's slide 2, retained here)

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Data processing pipeline for LLM pretraining

○ Modern LLMs (e.g., ChatGPT, Gemini, Claude, DeepSeek,...) employ the following general data pipeline:

1. Data cleaning

- ▶ Deduplication
- ▶ Privacy and copyright
- ▶ Quality filtering
- ▶ Data age

2. Data selection

- ▶ Data masking
- ▶ Document packing
- ▶ Data ordering
- ▶ Synthetic data

3. Data mixing

- ▶ Find optimal domain mixtures
- ▶ Target evaluations
- ▶ Online mixing approaches
- ▶ Offline mixing approaches

Data cleaning: Deduplication

- Datasets curated from internet contain substantial amounts of duplicate and near-duplicate content.
- This redundancy can negatively impact the performance of language models [51].
 - ▶ Redundancy skews the model's learning, resulting in overemphasis on repeated patterns.
- Effective deduplication helps to create a more balanced and varied training set.
 - ▶ It helps with model's generalization and alignment.
 - ▶ It reduces storage, which in turn reduces costs associated with infrastructure.
- We discuss three deduplication techniques in the sequel:
 1. Exact matching
 2. Approximate matching
 3. Model-based matching

Data cleaning: Filtering low-quality data

- Pretraining datasets for LLMs often contain substantial amounts of low-quality, noisy, or undesirable content.
- Modern LLMs use quality and/or toxicity filtering to achieve optimal model performance and desired behavior.

Example

Consider the Common Crawl dataset, a massive web scrape. It includes

- ▶ advertisements and promotional material;
- ▶ repetitive, non-human-readable text;
- ▶ toxic content.

Filtering approaches

Two primary approaches are used, often in combination:

1. Heuristic-based filtering: Rule-based methods that remove documents based on easily computable features.
2. Classifier-based filtering: Models trained to identify and remove specific types of undesirable content.

Data cleaning: Data age

- Language evolves, impacting the LLM performance on temporally misaligned data [59].
 - ▶ Temporal misalignment: Discrepancy between the time periods in training and evaluation [50, 60, 4].

Example

An LLM trained on 2019 data may perform poorly on a 2023 evaluation set containing information about events happened after 2019. Similarly, a model trained on 2023 data may struggle with nuances of language and information relevant to a 2019 evaluation set.

- Remarks:**
- Performance degradation is pronounced when the evaluation year is *after* the pretraining year [60].
 - Larger models exhibit greater sensitivity to temporal misalignment than smaller models.
 - Temporal misalignment complicates evaluation.
 - ▶ Older evaluations may underestimate newer models.
 - ▶ Newer evaluations may underestimate older models w.r.t. their original context.

Data mixing problem

- Pretraining an LLM requires large-scale data from diverse sources, such as the Pile and SlimPajama [29, 73].

Data Source	Sampling (%)	Train Tokens	Validation Tokens	Train (% of Total)
CommonCrawl	67.0	200.82B	214.72M	66.99
C4	15.0	44.98B	48.06M	15.00
GitHub	4.5	13.51B	14.42M	4.51
Books	4.5	13.49B	14.39M	4.50
Wikipedia	4.5	13.50B	14.41M	4.50
ArXiv	2.5	7.49B	8.01M	2.50
StackExchange	2.0	5.99B	6.41M	2.00
Total	100.0	299.78B	320.42M	100.00

Table: SlimPajama dataset [73]

- Denote k domains for pretraining as $D_{\text{train}} = \{D_1, D_2, \dots, D_k\}$.
- The sampling distribution is $\alpha = (\alpha_1, \dots, \alpha_k) \in \Delta^k$, the probability simplex over k domains.
- How to find the optimal data mixture weight α ?

How a trillion pages get cleaned without anyone reading them

STEP	THE IDEA, IN ONE SENTENCE	EVERYDAY ANALOGY
Exact deduplication	Compute a fingerprint (hash) of every page; identical fingerprints mean identical pages, keep one.	two photocopies of the same form
Near-duplicate detection	Chop each page into overlapping word windows, keep a few random samples as a signature, and compare signatures instead of pages.	recognising a reprint by a handful of sentences
Rule-based quality filters	Drop pages that are too short, mostly symbols, full of "..." lines, or without ordinary words like "the" and "and".	a spam folder's first rules
Classifier filters	Train a small model to recognise text that "looks like Wikipedia or a textbook" and keep only what it likes.	a reader hired to mark what is worth keeping

- The rules were written for English: a Thai page has no spaces to count words and no "the" to find. **Every filter needs a Thai version or it silently deletes Thai.**
- Add privacy filtering (phone numbers, emails, ID numbers), a date check (stale pages teach stale facts), and a recipe for how much of each source to use, and you have the pipeline behind Dolma, FineWeb, and every open model.

What deduplication and filtering actually bought

3.04%

of C4 examples are near-duplicates; one 61-word English sentence appears over 60,000 times (Lee et al., ACL 2022)

10x

less memorized text emitted after deduplication; train-test overlap affected over 4% of a standard validation set (Lee et al. 2022)

92%

of FineWeb removed by an educational-quality classifier at threshold 3 or above to make FineWeb-Edu, 1.3T tokens (Penedo et al., NeurIPS 2024)

33 → 37

MMLU after the FineWeb-Edu filter, ARC 46 to 57, in the dataset authors' own ablations across 70+ models

- Read the gains with the course's standing caveat: these are the **dataset authors' own ablations**, strong but not independent (section 2.4). DCLM reaches similar conclusions with a different classifier (fastText on OpenHermes and ELI5).
- The recurring finding across FineWeb-Edu, DCLM, and the mixing literature is **quality over quantity**: throwing away most of the web made the model better, not worse.
- Deduplication is also a **privacy** intervention: Kandpal et al. (ICML 2022) link duplicate counts to extraction risk. One fix, three benefits.

Contamination, ownership, and what "quality" quietly excludes

Your test set may be in the training data; the training data may be someone's copyrighted book; and the quality classifier prefers wealthy suburbs. Three audits every curation pipeline owes its users.

SOURCE DECKS

Lecture 11: Evaluation (CS224N Winter 2026, Stanford) · **Lecture 5: Data, Data, and Data**, ownership and bias sections (CS 329X, William Held, Stanford).

web.stanford.edu/class/cs224n · web.stanford.edu/class/cs329x © the respective authors · shown unmodified · selected pages

Benchmarks can be hard to trust for pretrained models



Horace He
@CHHillee

I suspect GPT-4's performance is influenced by data contamination, at least on Codeforces.

Of the easiest problems on Codeforces, it solved 10/10 pre-2021 problems and 0/10 recent problems.

This strongly points to contamination.

1/4

g's Race	implementation, math	🚩 ⭐	greedy, implementation	🚩 ⭐
nd Chocolate	implementation, math	🚩 ⭐	Cat?	implementation, strings
triangle!	brute force, geometry, math	🚩 ⭐	Actions	data structures, greedy, implementation, math
	greedy, implementation, math	🚩 ⭐	Interview Problem	brute force, implementation, strings

...



Susan Zhang ✓
@suchenzang

I think Phi-1.5 trained on the benchmarks. Particularly, GSM8K.



Susan Zhang ✓ @suchenzang · Sep 12
Let's take [github.com/openai/grade-s...](https://github.com/openai/grade-school-math)

If you truncate and feed this question into Phi-1.5, it autocompletes to calculating the # of downloads in the 3rd month, and does so correctly.

Change the number a bit, and it answers correctly as well.

1/ 🤖



Closed models + pretraining: hard to know that benchmarks are truly 'new'

Data De-contamination!

- Why does this happen?
 - Why internet data can be contaminated?
 - Why synthetic data can be contaminated?
- Data de-contamination practice
 - N-gram overlap: check for exact or near-exact n-gram matches (commonly 8-13 grams) between training data and benchmark examples
 - Sometimes mixed with embedding-based or paraphrase-based near-duplicate detection

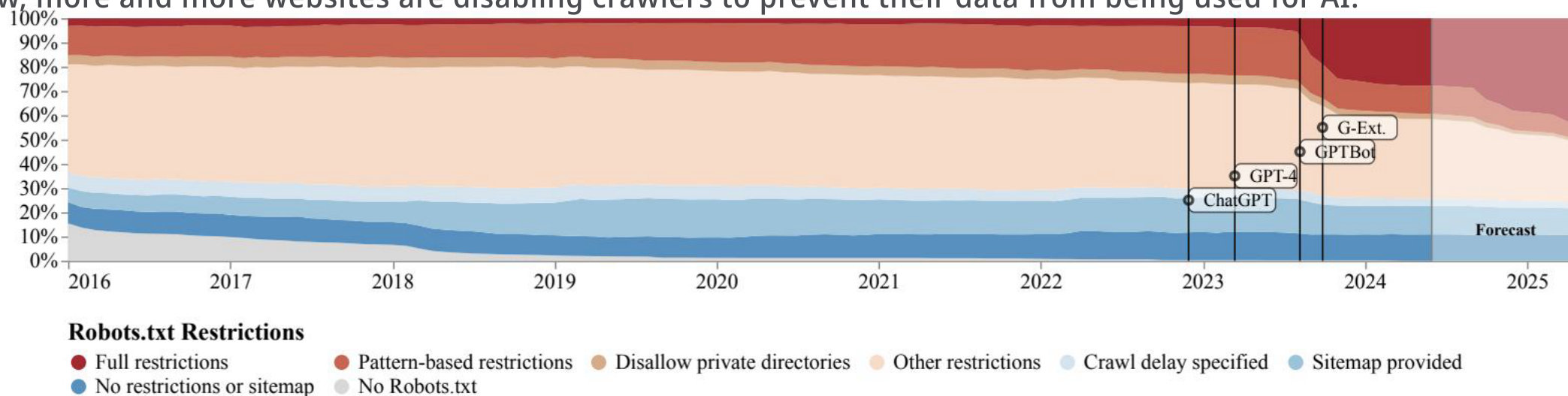


Copyright is not the only data ownership mechanism

Robots.txt is a file websites can host at their root that defines their rules for how web-scrapers should access them.

Before LLMs, the vast majority of websites had relatively permissive scraping rules so that they could be indexed by search engines!

Now, more and more websites are disabling crawlers to prevent their data from being used for AI.



Longpre et al. 2024

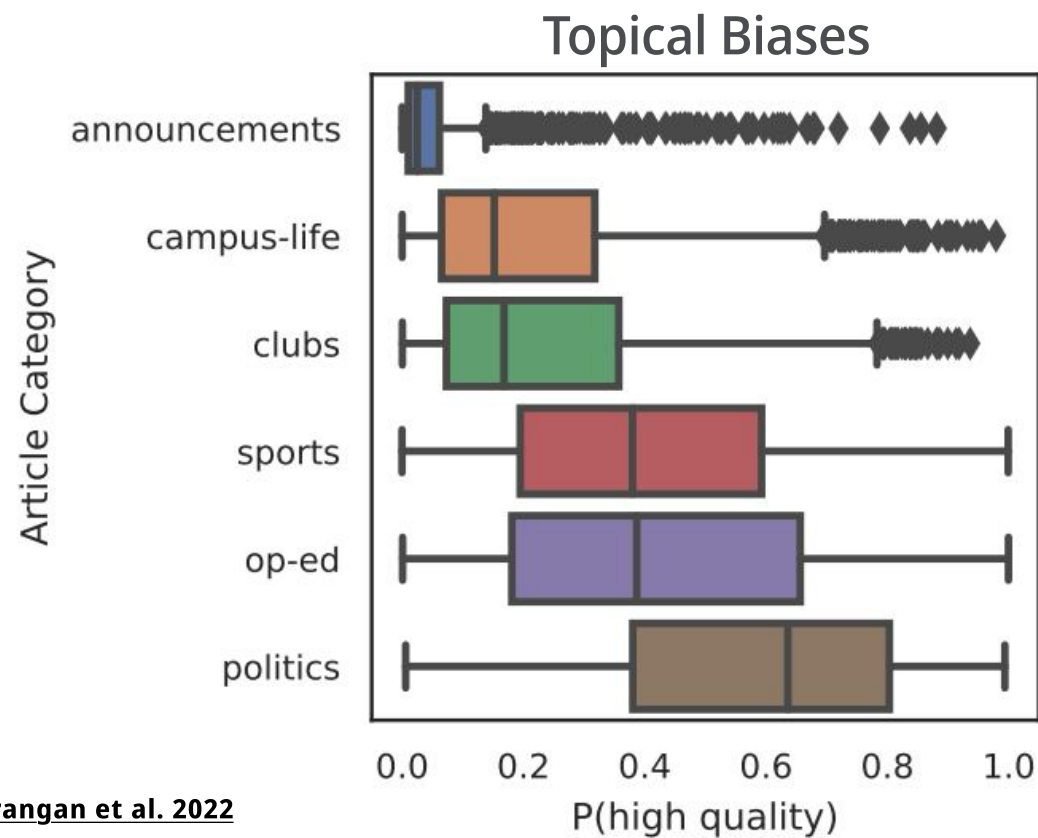
Topical Bias in Quality Classification

Top Source Domains

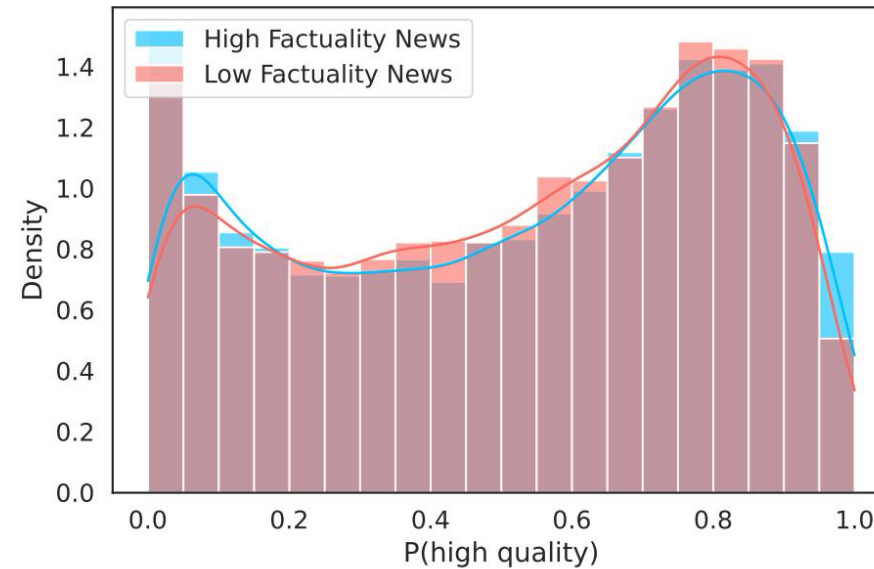
daily mail.co.uk
go.com

Table 1: The most p
in OpenWebText. M
whelming majority of

Gururangan et al. 2022



What does this definition of “quality” end up capturing?



Gururangan et al. 2022

Instruction data: the smallest dataset that matters most

From 52k machine-written Alpaca pairs to 1,000 hand-picked LIMA examples: how instruction data is generated, formatted, filtered, and who writes the labels.

SOURCE DECKS

Lecture 8: Post-training (CS224N Winter 2026, Stanford) · **Lecture 4: Fine-tuning**, instruction tuning and SFT datasets sections, EE-628 Training LLMs, EPFL (Cevher and Gulcehre).

web.stanford.edu/class/cs224n · epfl.ch/labs/lions/teaching/ee-628-training-large-language-models/ CS224N © the authors, shown unmodified · EPFL deck CC BY-NC-SA

Language modeling $\neq$ assisting users

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

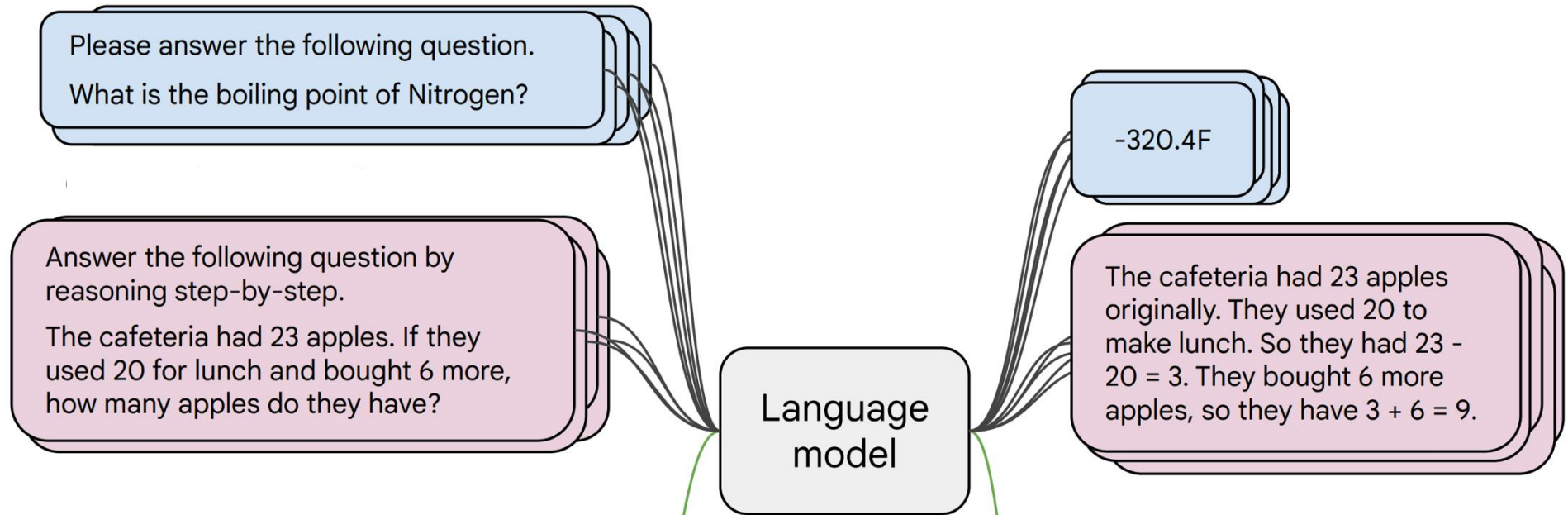
Explain evolution to a 6 year old.

Language models are not *aligned* with user intent [[Ouyang et al., 2022](#)]

Finetuning to the rescue!

Instruction finetuning

- **Collect examples** of (instruction, output) pairs across many tasks and finetune an LM



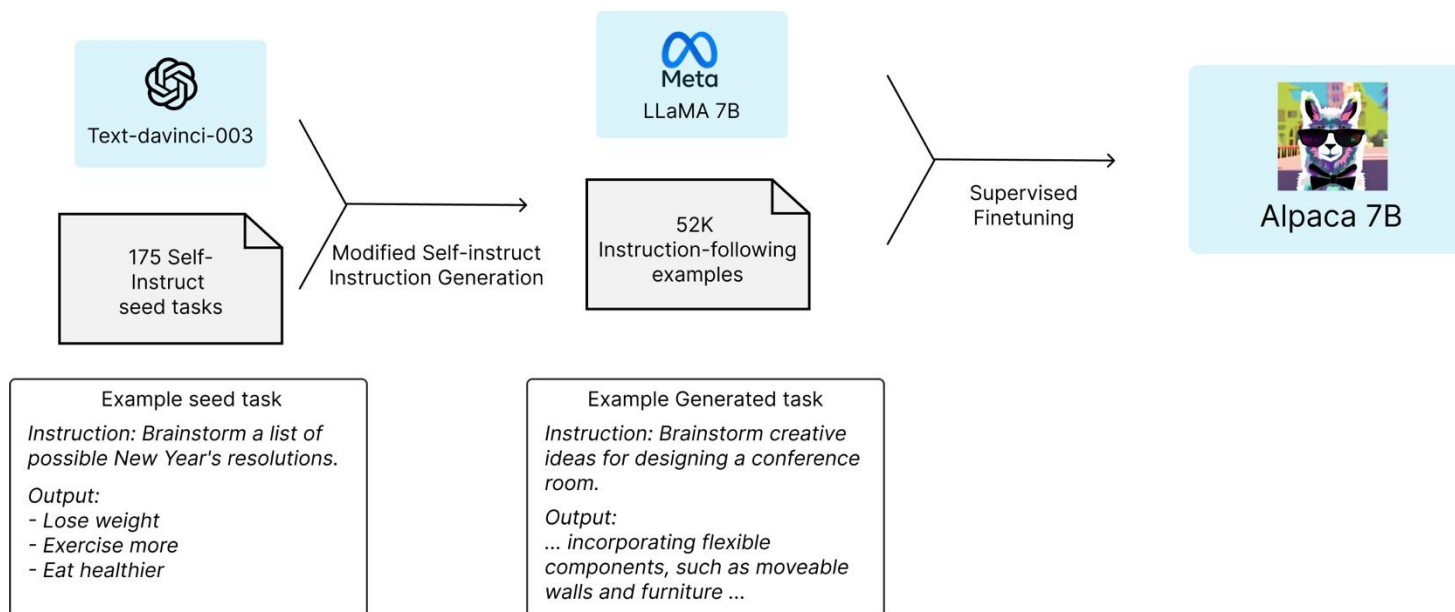
- Evaluate on **unseen tasks**



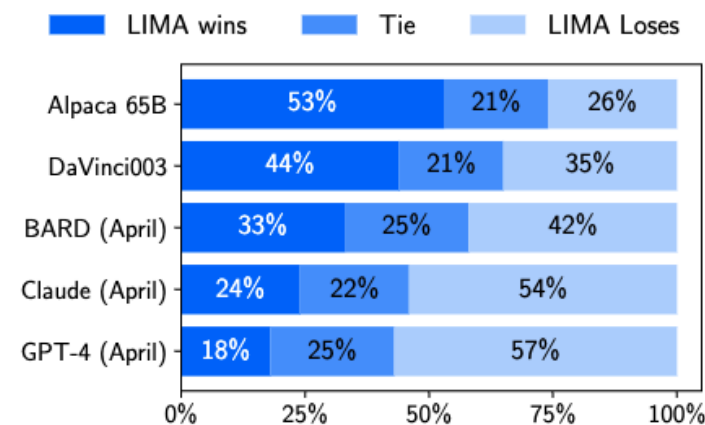
[FLAN-T5; [Chung et al., 2022](#)]

What have we learned from this?

- Generate instructions, input, and output from a LM [\[Wang et al., 2022\]](#)
- **Alpaca**: fine-tuned from the LLaMA 7B model on 52K instruction-following examples
- You don't need many samples to instruction tune (e.g., “*LIMA: Less Is More for Alignment*” [Zhou et al., 2023](#))



Source	#Examples
Training	
Stack Exchange (STEM)	200
Stack Exchange (Other)	200
wikiHow	200
Pushshift r/WritingPrompts	150
Natural Instructions	50
Paper Authors (Group A)	200



Where does the labels come from?

BUSINESS • TECHNOLOGY
Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic
15 MINUTE READ



SIAMH ROME BUSINESS 15.10.2023 00:00 AM
Millions of Workers Are Training AI Models for Pennies
From the Philippines to Colombia, low-paid workers label training data for AI models used by the likes of Amazon, Facebook, Google, and Microsoft.



Oskana Vero Fuentes with her dog. COURTESY OF OSKANA VERO FUENTES

Behind the AI boom, an army of overseas workers in 'digital sweatshops'

By Rebecca Tan and Baseline Caballo
August 28, 2023 at 2:00 a.m. EDT

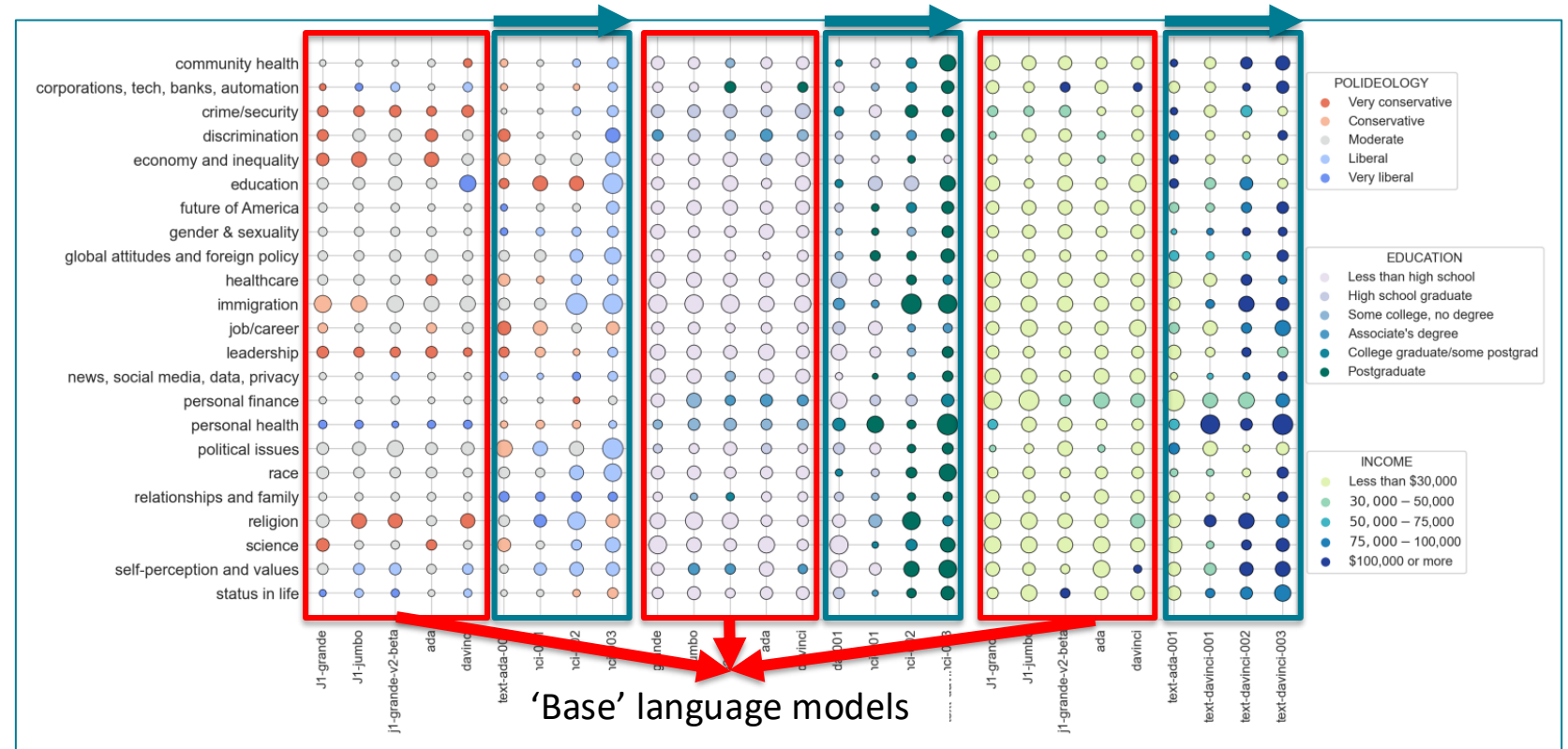


- RLHF labels are often obtained from overseas, low-wage workers

Where does the label come from?

Table 12: Labeler demographic data

What gender do you identify as?	
Male	50.0%
Female	44.4%
Nonbinary / other	5.6%
What ethnicities do you identify as?	
White / Caucasian	31.6%
Southeast Asian	52.6%
Indigenous / Native American / Alaskan Native	0.0%
East Asian	5.3%
Middle Eastern	0.0%
Latinx	15.8%
Black / of African descent	10.5%
What is your nationality?	
Filipino	22%
Bangladeshi	22%
American	17%
Albanian	5%
Brazilian	5%
Canadian	5%
Colombian	5%
Indian	5%
Uruguayan	5%
Zimbabwean	5%
What is your age?	
18-24	26.3%
25-34	47.4%
35-44	10.5%
45-54	10.5%
55-64	5.3%
65+	0%
What is your highest attained level of education?	
Less than high school degree	0%
High school degree	10.5%
Undergraduate degree	52.6%
Master's degree	36.8%
Doctorate degree	0%



[Santurkar+ 2023, OpinionQA]

- We also need to be quite careful about how annotator biases might creep into LMs

*Frame **all** tasks in the form of
natural language instruction to natural language response mapping*



Alpaca - An Example

```
{  
  "instruction": "Create a classification task by clustering the given list of items.",  
  "input": "Apples, oranges, bananas, strawberries, pineapples",  
  "output": "Class 1: Apples, Oranges\nClass 2: Bananas, Strawberries\nClass 3: Pineapples",  
  "text": "Below is an instruction that describes a task, paired with an input that provides further context."  
}
```

- *instruction*: describes the task the model should perform. Each of the 52K instructions is unique.
- *input*: optional context or input for the task. For example, when the instruction is "Summarize the following article", the input is the article. Around 40% of the examples have an input.
- *output*: the answer to the instruction as generated by text-davinci-003.
- *text*: the instruction, input and output formatted with the [prompt template](#) used by the authors for fine-tuning their models.

Augment with a model, but keep the humans in the loop

```
# LLM augmentation for a Thai clinical classifier
prompt = f"""Rewrite this note in the same clinical
register, keep every finding and the diagnosis,
change wording only. Note: {note}"""
para = llm(prompt)
if keeps_label(para, note):      # judge or human
    augmented.append((para, label))

# provenance, always
record = {"text": para, "source": "synthetic",
          "generator": MODEL_ID, "seed": note_id}
```

- Self-Instruct and Alpaca showed that a model can write its own training data; Evol-Instruct showed it can make that data harder. The **filtering step** is what separates a dataset from noise.
- **Model collapse:** Shumailov et al., Nature 2024 (cite this, not the earlier preprint): models trained recursively on generated data lose the tails of the distribution, generation after generation.
- Rules of the road: mark synthetic rows, cap their share, never evaluate on them, and keep a fresh human-written slice for validation.
- For Thai, generated text needs a native reader: fluency errors that an English judge model misses are common.

RAG data preparation: chunks are the new documents

Retrieval-augmented generation moves preprocessing to inference time: how you split, embed, and index documents decides what the model can ever cite.

SOURCE DECKS

Information Retrieval and RAG, RAG section (SLP3 slides, Jurafsky and Martin, Nov 2025) · **Lecture 10: RAG and agents** (CS224N Winter 2026, Stanford).

web.stanford.edu/~jurafsky/slp3/slides · web.stanford.edu/class/cs224n · © the respective authors · shown unmodified · selected pages

LLMs Hallucinate

Hallucination: a response that is not faithful to the facts of the world.

In the legal domain LLMs were shown to hallucinate up to 88% of the time!

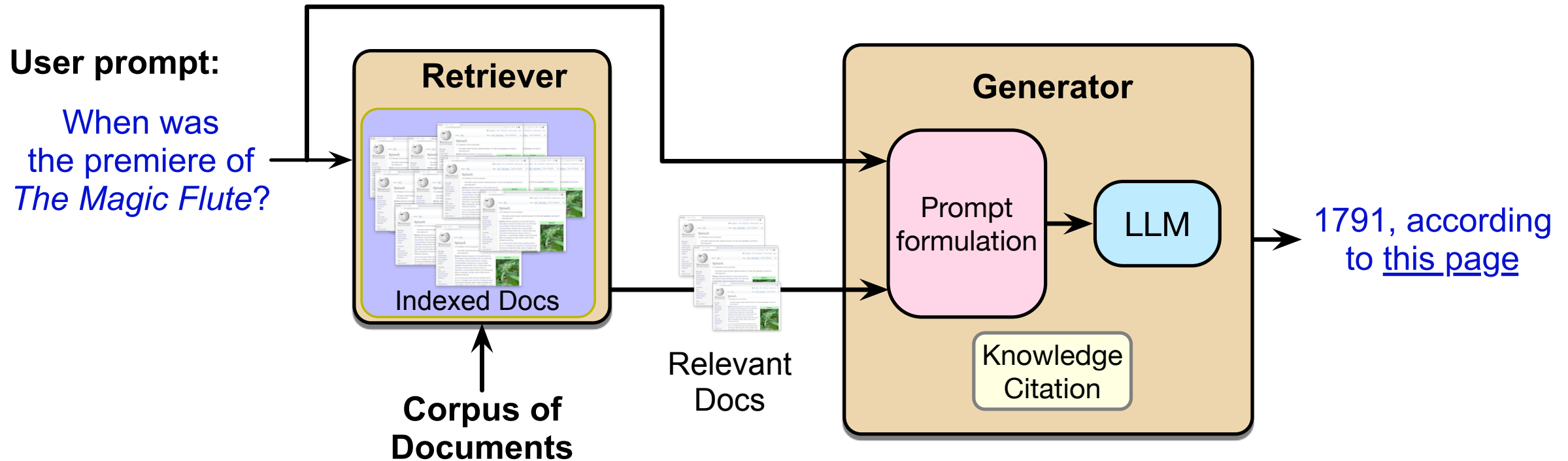
Dahl et al. (2024)

Solution: RAG

Retrieval-Augmented Generation

1. Use IR to **retrieve** documents from some collection
2. Then use LLM to **generate** an answer conditioned on the documents

Retrieval Augmented Generation (RAG)



Basic RAG

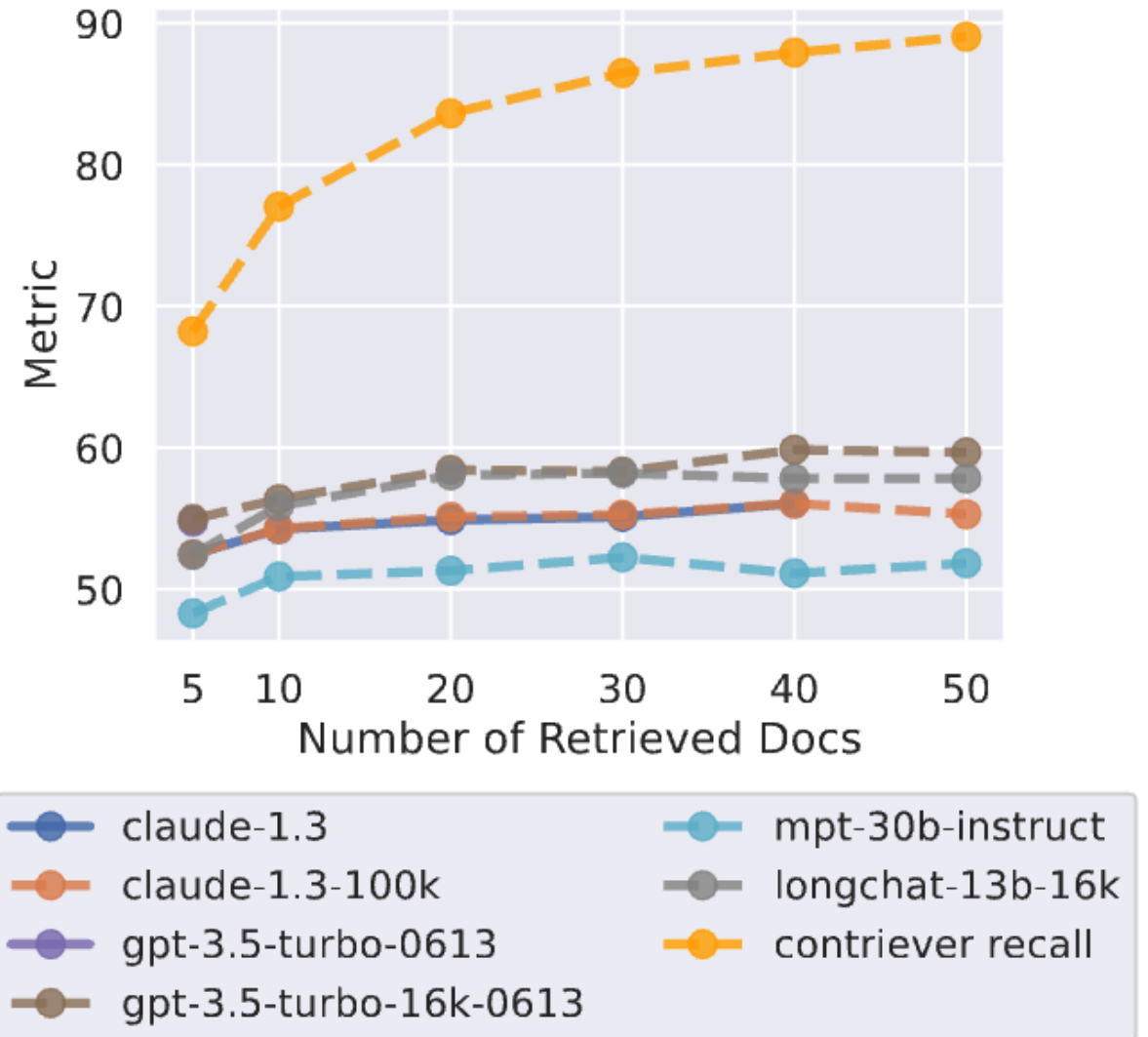
Given a document collection D and a user query q

- Call a retriever to return top k passages
- Create a prompt that includes q and the passages
- Call an LLM with the prompt

In practice: LMs cant use many documents

Retriever performance (yellow)
rises slowly to 90% recall

RAG performance (other lines)
saturate very quickly – after 10-
20 documents.



Before a model can cite a document, someone has to cut it up

- A retrieval system does not store whole documents; it stores **chunks** of a few hundred tokens, each turned into a vector so that similar passages can be found quickly.
- **Too small** and a chunk loses its context ("it was elevated" without saying what); **too large** and the relevant sentence drowns in unrelated text, and fewer chunks fit into the prompt.
- The usual recipe: split at paragraph breaks, then at sentences, then at spaces, with a small **overlap** so a sentence on a boundary is not lost.
- **Thai has no full stops:** a Thai sentence splitter must be used, or chunks break in the middle of a clause.
- Keep, with every chunk, **where it came from:** document, page, section. That is what makes a citation possible and lets you remove a document later.
- Newer tricks add context to each chunk before storing it; the proposers report large gains, and independent work finds that clever splitting does not always pay. **Test on real questions** from your users before choosing.

Clinical text, learned era: context, features, de-identification

Mr. Huntington was treated for Huntington's disease at Huntington Hospital. Why context beats term lists, how learned de-identification works, and what changes when the model is a transformer.

SOURCE DECK

Lecture 8: Natural Language Processing, Part 2, 6.S897 / HST.956 Machine Learning for Healthcare, Spring 2019 · Peter Szolovits, MIT OpenCourseWare.

<https://ocw.mit.edu/courses/6-s897-machine-learning-for-healthcare-spring-2019/> License: CC BY-NC-SA 4.0 · attribution page retained · third-party figures marked in the deck stay under their own terms

The Importance of Context

- “Mr. **Huntington** was treated for **Huntington’s** Disease at **Huntington** Hospital, located on **Huntington** Avenue.”
 - Huntington
 - Huntington’s Disease
 - Mr. Huntington’s Disease
- “Atenolol was **administered** to Mr. Huntington.”
 - vs. “Atenolol was **considered** for control of heart rate.”
 - vs. “Atenolol was **ineffective** and therefore **discontinued**.”

Using this model for de-identification

Courtesy of Elsevier, Inc., <https://www.sciencedirect.com>.
Used with permission.

Uzuner, Ö., Sibanda, T. C., Luo, Y., & Szolovits, P. (2008). A de-identifier for medical discharge summaries. *Artificial Intelligence in Medicine*, 42(1), 13–35. <http://doi.org/10.1016/j.artmed.2007.10.001>

Table 6 Evaluation on authentic discharge summaries

Method	Class	Precision (%)	Recall (%)	F-measure (%)
Stat De-id	PHI	98.46	95.24	96.82
IFinder	PHI	26.17	61.98	36.80 *
H + D	PHI	82.67	87.30	84.92 *
CRFD	PHI	91.16	84.75	87.83 *
Stat De-id	Non-PHI	99.84	99.95	99.90
IFinder	Non-PHI	98.68	94.19	96.38 *
H + D	Non-PHI	99.58	99.39	99.48 *
CRFD	Non-PHI	99.62	99.86	99.74 *

The *F*-measure differences from Stat De-id in PHI and in non-PHI are significant at $\alpha = 0.05$.

Table 7 Evaluation of SNoW and Stat De-id on authentic discharge summaries

Method	Class	Precision (%)	Recall (%)	F-measure (%)
Stat De-id	PHI	98.40	93.75	96.02
SNoW	PHI	96.36	91.03	93.62 *
Stat De-id	Non-PHI	99.90	99.98	99.94
SNoW	Non-PHI	99.86	99.95	99.90 *

The *F*-measure differences from Stat De-id in PHI and in non-PHI are significant at $\alpha = 0.05$.

“Revolutionary Advances” in Embeddings

- The year 2018 has been an inflection point for machine learning models handling text (or more accurately, Natural Language Processing or NLP for short). Our conceptual understanding of how best to represent words and sentences in a way that best captures underlying meanings and relationships is rapidly evolving.
—Jay Alammar (<http://jalammar.github.io/illustrated-bert/> — *good tutorial*)
- Bidirectional LSTM applied to learn context-specific embeddings (ELMo)
- Transformer architecture — focus on attention mechanism
- Bidirectional Encoder Representations from Transformers (BERT)
- Generative Pre-Training (GPT-2) — transformer with multi-task training, huge corpus, huge model

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6.S897 / HST.956 Machine Learning for Healthcare

Spring 2019

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From six rules to a model that reads the note

- Class 1 used **rules**: patterns for dates, phone numbers, ID numbers, emails. They are transparent and easy to audit, and they miss names hidden inside sentences.
- The learned approach treats de-identification as **labelling every word**: name, date, location, identifier, or nothing. Trained on de-identified hospital corpora (the i2b2 and n2c2 challenges), it catches what rules cannot.
- Learned systems also **miss unpredictably**, so the safe recipe is model first, rules as a safety net, then a human reads a random sample and reports what slipped through.
- A de-identified corpus is still **training data for a model that memorises**: what goes in can be extracted later. Deduplication and access rules are part of privacy, not separate from it.
- Reference corpora such as MIMIC come with data-use agreements; **signing them is a preprocessing step** too.
- Models trained on English notes **do not transfer** to Thai names, Thai ID formats, or Buddhist-era dates without Thai-labelled data.

What changes for Thai notes, and what does not

- **Subword tokenizers absorb code switching:** Thai narrative and English abbreviations pass through one vocabulary, no segmenter required. The classical segmentation problem largely dissolves at the model boundary.
- **The premium bites hardest here:** a long Thai discharge summary can cost 2 to 4 times the tokens of its English equivalent; prompt and context budgets must be set per language (part D2).
- **De-identification models trained on English i2b2 notes do not transfer:** Thai names, Thai ID formats, Buddhist-era dates, and hospital names need Thai-labelled data or the regex safety net from class 1.
- **Legal frame:** Thailand's Personal Data Protection Act (PDPA, B.E. 2562) treats health data as sensitive personal data; de-identification and consent are preprocessing requirements, not afterthoughts, and sending raw notes to an external API is a data transfer decision.
- **Local and open-weight models** keep data on site; measure their Thai fertility and Thai quality before assuming they match English performance.
- The audit habit from image class 2 carries over: sample, read, count errors, and record the tokenizer, model, and de-identification recall in the datasheet.

Regulatory details change; verify the current PDPA guidance and your hospital's data-governance rules before deployment.

If you remember nothing else

- 1. The tokenizer is part of the dataset: it fixes the vocabulary, the context budget, and the bill, and it charges Thai two to four times what it charges English.
- 2. Web-scale preprocessing is subtraction: deduplicate at three levels, filter with rules you can read and classifiers you have audited, and keep the numbers.
- 3. Every curation statistic and every selection score is fitted on training candidates only; benchmarks and validation data never influence them.
- 4. Quality filters, judge models, and synthetic data all have a direction of bias; check who and which language is being removed before celebrating the gain.
- 5. For clinical and Thai text, preprocessing is also compliance: de-identification recall, PDPA, and provenance belong in the methods section next to chunk size and tokenizer version.

Three things to try, no programming required

- **See the token tax.** Open the Tiktokenizer web demo (tiktokenizer.vercel.app), paste one paragraph in English and its Thai translation, and compare the token counts with two different models. Write down the ratio.
- **Read a data section.** Pick one open model (for example Llama 3 or OLMo) and find, in its paper or model card, how many tokens it was trained on, which sources, and what was filtered out. Note what is **not** disclosed.
- **Find a duplicate.** Take any text collection you have (emails, reports, notes) and look for two documents that say the same thing in nearly the same words. Decide whether a model should see both, and why.
- **Reading, optional.** The FineWeb blog post from Hugging Face (2024) explains an entire cleaning pipeline in plain language; Lee et al. (ACL 2022) is the short paper on duplicates.

Source decks used in class 2

- 1 **CS224N Natural Language Processing with Deep Learning, Winter 2026, Stanford University.** Lecture 14, Tokenization and Multilinguality (guest lecture by Julie Kallini); Lecture 7, Pretraining; Lecture 8, Post-training; Lecture 10, RAG and agents; Lecture 11, Evaluation.
<https://web.stanford.edu/class/cs224n/> · © the authors; shown unmodified with attribution; no explicit reuse license.

- 2 **EE-628 Training Large Language Models, Spring 2025, EPFL (LIONS),** Volkan Cevher and Caglar Gulcehre. Lecture 3, Data; Lecture 4, Fine-tuning.
<https://www.epfl.ch/labs/lions/teaching/ee-628-training-large-language-models/> · Creative Commons BY-NC-SA, as stated on each deck's license slide (retained).

- 3 **CS 329X Human-Centered LLMs, Stanford University,** Lecture 5, Data, Data, and Data (William Held).
<https://web.stanford.edu/class/cs329x/> · © the author; shown unmodified with attribution.

- 4 **Speech and Language Processing, 3rd edition draft, lecture slides,** Dan Jurafsky and James H. Martin: Large Language Models introduction (Aug 2026 version, scaling and pretraining-data pages), Words and Tokens (BPE section), Large Language Models (Aug 2025), Information Retrieval and RAG (Nov 2025).
<https://web.stanford.edu/~jurafsky/slp3/slides/> · © the authors, posted freely for teaching; shown unmodified.

- 5 **MIT OpenCourseWare, 6.S897 / HST.956 Machine Learning for Healthcare, Spring 2019,** Lecture 8, NLP Part 2 (Peter Szolovits). CC BY-NC-SA 4.0; OCW attribution page retained; marked third-party figures excluded from the license.

Demos, papers, and further reading

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- 6 **Authored slides** (marked in the footer) were written for this course. Live measurements used tiktoken (cl100k_base, o200k_base) and the XLM-R SentencePiece tokenizer via Hugging Face tokenizers; translations of the demo sentence were written for this deck. The code-level demos (BPE training, MinHash, quality rules, contamination check, chunking) live in the extended version of this deck. Training-token figures on the D0 chart are as reported in the respective model papers and cards (BERT, GPT-3, Gopher, PaLM, Chinchilla, LLaMA, Llama 2, Llama 3, Qwen2.5).
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- 7 **Papers cited on authored slides:** Petrov et al. NeurIPS 2023; Ahia et al. EMNLP 2023; Xue et al. ByT5 TACL 2022; Clark et al. CANINE TACL 2022; Lee et al. ACL 2022; Kandpal et al. ICML 2022; Penedo et al. FineWeb NeurIPS 2024; Rae et al. Gopher 2021; Zhou et al. LIMA NeurIPS 2023; Chen et al. AlpaGasus ICLR 2024; Liu et al. DEITA ICLR 2024; Xia et al. LESS ICML 2024; Li et al. IFD NAACL 2024; Shumailov et al. Nature 2024; Gunther et al. Late Chunking 2024; Anthropic Contextual Retrieval 2024; Qu, Tu and Bao, Findings of NAACL 2025; Kapoor and Narayanan Patterns 2023; Johnson et al. MIMIC Scientific Data 2016.
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- 8 **Not slides, but recommended:** Stanford CS336 Language Modeling from Scratch, executable data lectures (stanford-cs336.github.io) · Hugging Face LLM course, chapter on tokenizers (Apache-2.0) · Dolma and FineWeb dataset papers for published filtering rules.
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- 9 **Compilation note.** Curated by T. Intharah for the Research Master's module Data Preparation for ML and AI, Khon Kaen University, September 2026. For classroom use; each source retains its own license.